

```
`timescale 1ns / 1ps
```

```
module traffic_controller_tb;
```

```
    reg clk;
```

```
    reg rst;
```

```
    wire [1:0] lights_A;
```

```
    wire [1:0] lights_B;
```

```
    traffic_controller uut (
```

```
        .clk(clk),
```

```
        .rst(rst),
```

```
        .lights_A(lights_A),
```

```
        .lights_B(lights_B)
```

```
    );
```

```
    // Generate clock
```

```
    initial begin
```

```
        clk = 0;
```

```
        forever #5 clk = ~clk;    // 10ns period
```

```
    end
```

```
    // Stimulus
```

```
    initial begin
```

```
        rst = 1;
```

```
        #20;                // Hold reset high for 20ns
```

```
        rst = 0;
```

```
        #1000;              // Run for 1000ns
```

```
        $finish;
```

```
    end
```

```
endmodule
```